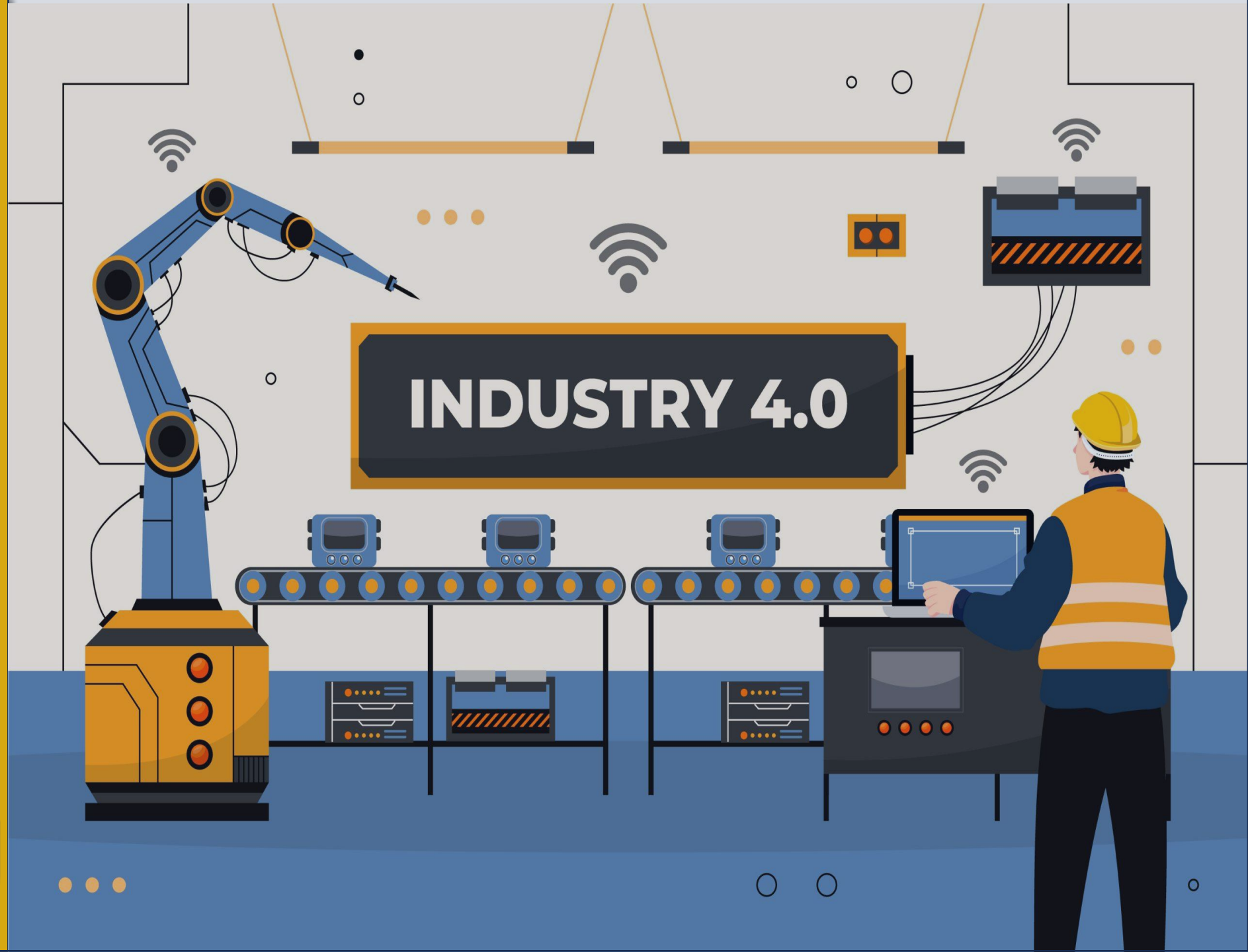
Smart Sensors & Sensor Fusion

Automation & Industry 4.0

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Agenda

Content of the Presentation

1. Introduction & Evolution
2. Inside the Smart Sensor and Fusion Systems
3. **Sensor Fusion Levels:** Data, Feature, and Decision level fusion
4. **Applications in Industry**
5. State of the Art & Wolrd Maket
6. Conclusion

Introduction

Smart Sensors & Sensor Fusion Definitions

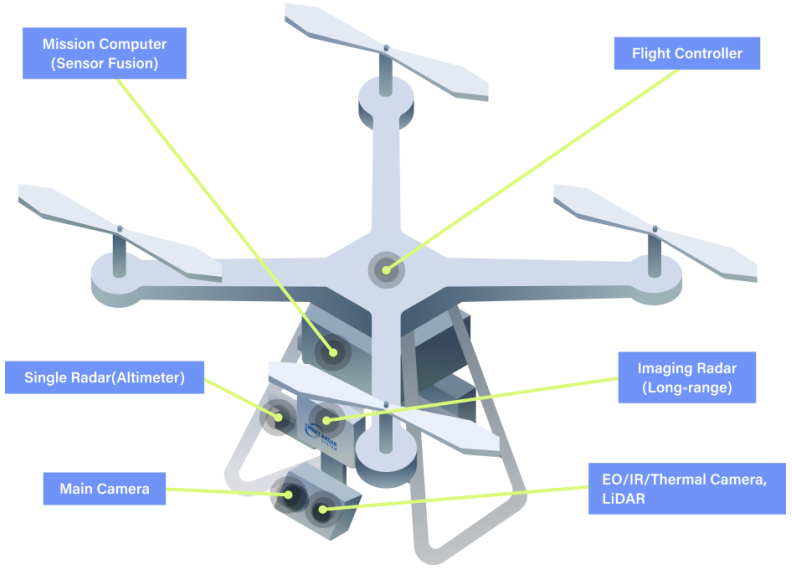
Smart Sensors:

Advanced devices that integrate a sensing element with embedded processing (microcontrollers) to collect, analyze, and transmit data locally, rather than just simple detection.



Sensor Fusion:

A multilevel process combining data from multiple sources to generate information that is more precise and reliable than any single sensor can achieve alone.



Introduction

The Evolution of Sensing

Industry 3.0 (Past):



- Technology: 4-20mA Analog Loops.
- Function: Unidirectional communication (One Wire = One Signal). Sensors were "passive" and isolate

Industry 4.0 (Present):



- Technology: IO-Link & Smart Sensors.
- Function: Bi-directional digital communication. Sensors perform self-diagnosis and dynamic configuration

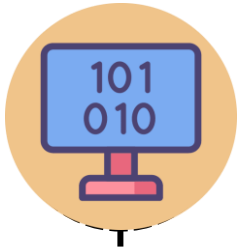
Future Outlook:

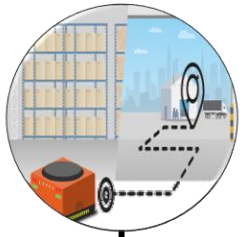


- Technology: Wireless IIoT & TinyML.
- Function: "Sensor-as-a-Service." Sensors act as intelligent nodes in a global network, using edge AI for cognitive sensing.

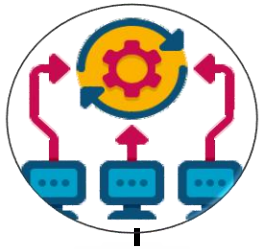
Architecture & Components

Sensor Fusion Levels & Architecture









Low level (Data Level)

Medium-Level (Feature Level)

High-Level (Decision Level)

Integration Architecture:

Combines raw data from similar sensors (e.g., visual + thermal pixels) to maximize information content.

Extracts specific features (edges, shapes, movement) from different sensors and fuses them. Used in robot navigation (SLAM).

Each sensor makes a local decision (vote), and the system combines them (e.g., voting "danger" for emergency stops).

Ranges from **Loosely Coupled** (independent solutions blended) to **Ultra-Tightly Coupled** (raw data fused at signal processing level).

Architecture & Components

Anatomy of a Smart Sensor

Primary Sensing Element (Transducer) :

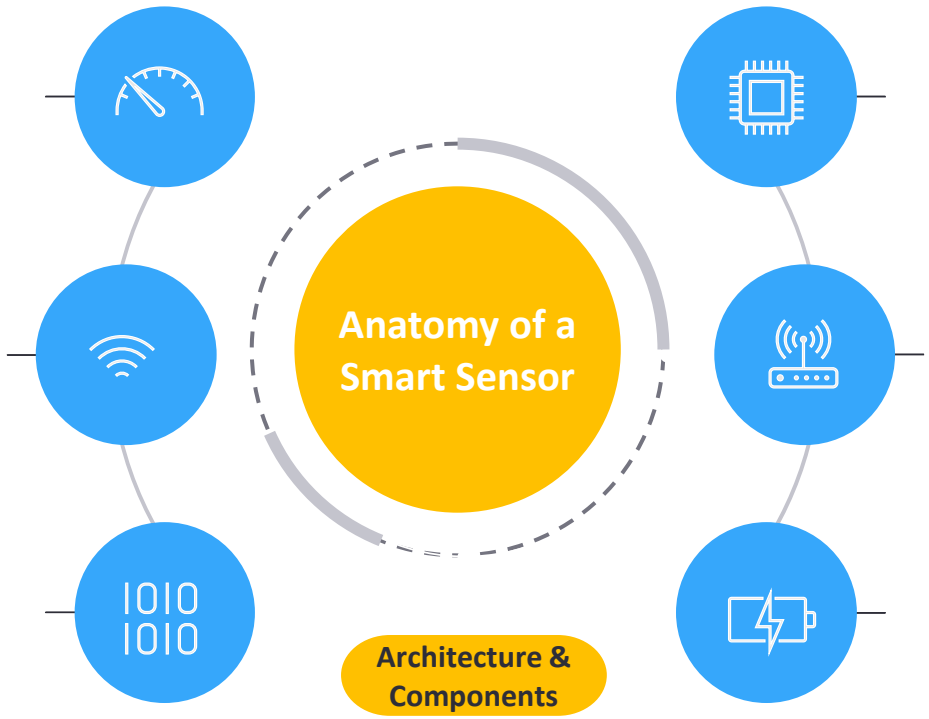
The interface with the physical world (e.g., MEMS). Converts physical quantities (heat, pressure) into raw electrical signals.

Signal Conditioning Unit (SCU):

Amplifies weak signals and filters out noise/interference to prepare the signal for digitization

Analog-to-Digital Converter (ADC):

The bridge between the analog and digital worlds. Converts voltage into binary code; its resolution determines sensor sensitivity.



Microcontroller Unit (MCU):

The "Brain" of the sensor. It executes firmware for digital filtering, self-calibration, and local data processing

Communication Transceiver:

Handles data transmission. Wired (IO-Link) or Wireless (Bluetooth LE/ZigBee) interfaces to connect to the IIoT network.

Power Management Unit (PMU):

Regulates voltage and manages energy harvesting (vibration/solar) for self-powered, "deploy-and-forget" operations.

Applications in the Industry

Indoor Positioning (Logistics & Navigation) - Feature Level

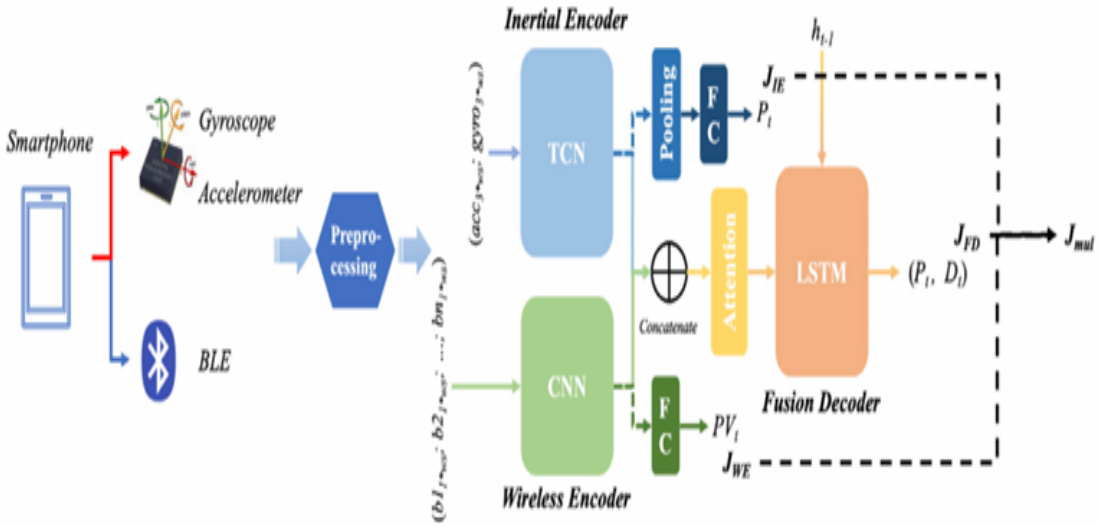
Problem Statement:

- GPS does not work indoors.
- **Bluetooth (BLE)** signals bounce off factory walls, creating noise.
- **Motion Sensors (IMU)** suffer from drift over time.
- *Result:* Using either technology alone leads to inaccurate tracking of robots and workers

The Solution (SensNet Architecture):

- **Feature Level Fusion:** Instead of calculating two different locations, the system uses Deep Learning (CNN + TCN) to extract abstract "features" from both noisy BLE signals and IMU movement patterns.

- **Fusion Decoder:** An LSTM network fuses these features into a single vector to calculate precise position.
- **Impact:** Achieves high precision with an error rate of only **0.415 meters**, far superior to individual sensors.



Applications in the Industry

Digital Twins (Siemens Amberg Electronics Plant)

Key Concepts:

- **Virtual Replica:** A precise digital copy of the factory floor capable of running "what-if" simulations (e.g., increasing conveyor speeds).

Sensor Fusion Role:

- **RFID Systems:** Identify 1,200+ product variants to trigger specific production programs.
- **Vision Sensors:** Fuse visual data with CAD models to detect microscopic defects.

Outcome :

The system handles ~50 million data points per day, achieving a **99.9988% quality rate**





State of the Art & Market

World Market

(Market Overview):

Market Size: Valued at ~$63 Billion in 2024, projected to reach $136 Billion by 2029 (CAGR ~17%).

Honeywell	Siemens AG	Robert Bosch (MEMS Leader)	Regional Dominance
~14.3% Share. Leader in industrial automation, specifically pressure and airflow sensors for process control.	**~12.7% Share.** Dominates "Digital Factory" solutions, integrating smart sensors directly into PLCs and Digital Twins.	**The #1 global** supplier of MEMS sensors (accelerometers/gyroscopes) crucial for logistics and wearables	**Asia Pacific (APAC)** holds the largest share (~40%) due to manufacturing hubs in China/India; **North America** leads in high-value innovation (Aerospace/Defense).

State of the Art & Market

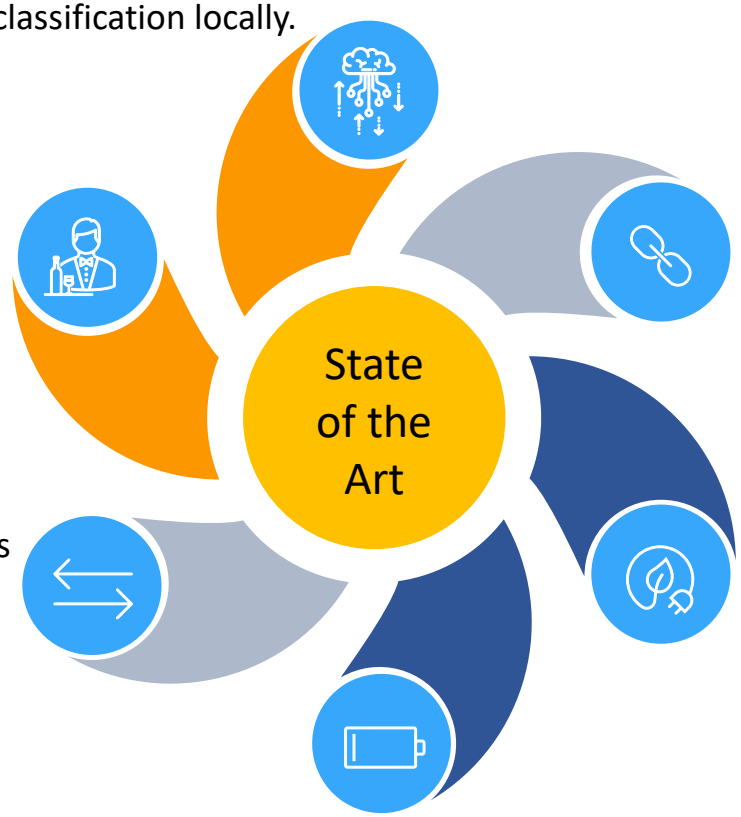
State of the Art Smart Sensors

Edge Intelligence:

By embedding AI algorithms, sensors can now perform complex pattern recognition and classification locally.

Sensor-as-a-Service:

Through Cloud-based IoT platforms, sensor data is instantly available globally, allowing for "virtualized" control where physical processes are monitored and optimized via digital twins



Deep Integration:

State-of-the-art fusion systems now employ Ultra-Tightly Coupled Integration (UTCI), where raw data from disparate are fused at the signal processing level.

Bi-directional Ecosystems:

The integration of standards like IO-Link Wireless has eliminated the final physical barrier (cables), means that data flows in both directions.

Energy Harvesting:

Smart sensors use energy harvesting to convert ambient energy, such as vibration, heat, or light into electrical power. Technologies like **Piezoelectric Nanogenerators (PENGs)** and **Triboelectric Nanogenerators (TENGs)** enable fully self-sustaining sensor operation

Low-Power Protocols: This is complemented by the adoption of ultra-low-power communication protocols (e.g., Bluetooth Low Energy, ZigBee, and LoRaWAN), ensuring that the harvested energy is sufficient for continuous operation

Conclusion

Pros & Cons

Advantages (Pros):

- **Precision:** Fusion eliminates blind spots (e.g., LiDAR + IMU) and improves accuracy via statistical weighting
- **Predictive Maintenance:** Detects faults (bearing wear) weeks in advance, reducing downtime.
- **Agility:** IO-Link allows instant remote parameter changes for flexible production.

Challenges (Cons):

- **Cost & Complexity:** High initial capital for smart nodes; requires specialized math/AI skills.
- **Data Overload:** Streaming raw data strains bandwidth (solved by Edge computing).

Conclusion:

- We are shifting from "Smart Sensing" to **"Cognitive Sensing"**.
- Future sensors will be autonomous, self-powered (energy harvesting), and use TinyML to process data at the edge, becoming the intelligent "sensory organs" of Industry 4.0.

